

- Many side effects are time-dependent (i.e., they start immediately upon dosing and upon each dose increase, but go away with time)
- Activation and agitation may represent the induction of a bipolar state, especially a mixed dysphoric bipolar II condition sometimes associated with suicidal ideation, and require the addition of lithium, a mood stabilizer or an atypical antipsychotic, and/or discontinuation of atomoxetine

Dosing and Use

Usual Dosage Range

- 0.5–1.2 mg/kg/day in children up to 70 kg; 40–100 mg/day in adults

Dosage Forms

- Capsule 10 mg, 18 mg, 25 mg, 40 mg, 60 mg, 80 mg, 100 mg

How to Dose

- For children 70 kg or less: initial dose 0.5 mg/kg per day; after 7 days can increase to 1.2 mg/kg per day either once in the morning or divided; maximum dose 1.4 mg/kg per day or 100 mg/day, whichever is less
- For adults and children over 70 kg: initial dose 40 mg/day; after 7 days can increase to 80 mg/day once in the morning or divided; after 2–4 weeks can increase to 100 mg/day if necessary; maximum daily dose 100 mg

Dosing Tips

- Can be given once a day in the morning
- * Efficacy with once daily dosing despite a half-life of 5 hours suggests therapeutic effects persist beyond direct pharmacologic effects, unlike stimulants whose effects are generally closely correlated with plasma drug levels
- Once daily dosing may increase gastrointestinal side effects
- Lower starting dose allows detection of those patients who may be especially sensitive to side effects such as tachycardia and increased blood pressure
- Patients especially sensitive to the side effects of atomoxetine may include those individuals deficient in the enzyme that metabolizes atomoxetine, CYP450 2D6 (i.e., 7% of Caucasians and 2% of African Americans)
- In such individuals, drug should be titrated slowly to tolerability and effectiveness
- Other individuals may require up to 1.8 mg/kg total daily dose

Overdose

- No fatalities have been reported as monotherapy; sedation, agitation, hyperactivity, abnormal behavior, gastrointestinal symptoms

Long-Term Use

- Safe

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Metabolized by CYP450 2D6
- Half-life approximately 5 hours
- Food does not affect absorption



Drug Interactions

- Tramadol increases the risk of seizures in patients taking an antidepressant
- Plasma concentrations of atomoxetine may be increased by drugs that inhibit CYP450 2D6 (e.g., paroxetine, fluoxetine), so atomoxetine dose may need to be reduced if coadministered
- Coadministration of atomoxetine and oral or IV albuterol may lead to increases in heart rate and blood pressure
- Coadministration with methylphenidate does not increase cardiovascular side effects beyond those seen with methylphenidate alone

- Use with caution with MAOIs, including 14 days after MAOIs are stopped (for the expert)

Other Warnings/Precautions

- Growth (height and weight) should be monitored during treatment with atomoxetine; for patients who are not growing or gaining weight satisfactorily, interruption of treatment should be considered
- Use with caution in patients with hypertension, tachycardia, cardiovascular disease, or cerebrovascular disease
- Use with caution in patients with bipolar disorder
- Use with caution in patients with urinary retention, benign prostatic hypertrophy
- Rare reports of hepatotoxicity; although causality has not been established, atomoxetine should be discontinued in patients who develop jaundice or other evidence of significant liver dysfunction
- Use with caution with antihypertensive drugs
- Increased risk of sudden death has been reported in children with structural cardiac abnormalities or other serious heart conditions
- When treating children, carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment and make sure to document this in the patient's chart
- Distribute the brochures provided by the FDA and the drug companies

- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such symptoms immediately
- Monitor patients for activation of suicidal ideation, especially children and adolescents

Do Not Use

- If patient is taking an MAOI (except as noted under Drug Interactions)
- If patient has pheochromocytoma or history of pheochromocytoma
- If patient has a severe cardiovascular disorder that might deteriorate with clinically important increases in heart rate and blood pressure
- If patient has angle-closure glaucoma
- If there is a proven allergy to atomoxetine

Special Populations

Renal Impairment

- Dose adjustment not generally necessary

Hepatic Impairment

- For patients with moderate liver impairment, dose should be reduced to 50% of normal dose

- For patients with severe liver impairment, dose should be reduced to 25% of normal dose

Cardiac Impairment

- Use with caution because atomoxetine can increase heart rate and blood pressure
- Do not use in patients with structural cardiac abnormalities

Elderly

- Some patients may tolerate lower doses better
- Reduction in the risk of suicidality with antidepressants compared to placebo in adults age 65 and older



Children and Adolescents

- Approved to treat ADHD in children over age 6
- Recommended target dose is 1.2 mg/kg per day
- Do not use in children with structural cardiac abnormalities or other serious cardiac problems
- Carefully weigh the risks and benefits of pharmacological treatment against the risks and benefits of nontreatment and make sure to document this in the patient's chart
- Monitor patients face-to-face regularly, particularly during the first several weeks of treatment

- Use with caution, observing for activation of known or unknown bipolar disorder and/or suicidal ideation, and inform parents or guardians of this risk so they can help observe child or adolescent patients

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Some animal studies have shown adverse effects
- Use in women of childbearing potential requires weighing potential benefits to the mother against potential risks to the fetus
- * For women of childbearing potential, atomoxetine should generally be discontinued before anticipated pregnancies

Breast Feeding

- Unknown if atomoxetine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommend either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- No known abuse potential

Potential Disadvantages

- May not act as rapidly as stimulants when initiating treatment in some patients

Primary Target Symptoms

- Concentration, attention span
- Motor hyperactivity
- Depressed mood



Pearls

- * Unlike stimulants approved for ADHD, atomoxetine does not have abuse potential and is not a scheduled substance
- * Despite its name as a selective norepinephrine reuptake inhibitor, atomoxetine enhances both dopamine and norepinephrine in frontal cortex, presumably accounting for its therapeutic actions on attention and concentration
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, atomoxetine can

increase dopamine as well as norepinephrine in this part of the brain, presumably causing therapeutic actions in ADHD

- Since dopamine is inactivated by dopamine reuptake in nucleus accumbens, which largely lacks norepinephrine transporters, atomoxetine does not increase dopamine in this part of the brain, presumably explaining why atomoxetine lacks abuse potential
- Atomoxetine's known mechanism of action as a selective norepinephrine reuptake inhibitor suggests its efficacy as an antidepressant
- Pro-noradrenergic actions may be theoretically useful for the treatment of chronic pain
- Atomoxetine's mechanism of action and its potential antidepressant actions suggest it has the potential to destabilize latent or undiagnosed bipolar disorder, similar to the known actions of proven antidepressants
- Thus, administer with caution to ADHD patients who may also have bipolar disorder
- Unlike stimulants, atomoxetine may not exacerbate tics in Tourette's syndrome patients with comorbid ADHD
- Urinary retention in men over 50 with borderline urine flow has been observed with other agents with potent norepinephrine reuptake blocking properties (e.g., reboxetine, milnacipran), so administer atomoxetine with caution to these patients

- Atomoxetine was originally called tomoxetine but the name was changed to avoid potential confusion with tamoxifen, which might lead to errors in drug dispensing

Suggested Reading

Garnock-Jones KP , Keating GM . Atomoxetine: a review of its use in attention-deficit hyperactivity disorder in children and adolescents . *Paediatr Drugs* 2009 ;**11** (3):203–26 .

Kelsey D , Sumner C , Casat C , et al. Once daily atomoxetine treatment for children with attention deficit hyperactivity behavior including an assessment of evening and morning behavior: a double-blind, placebo-controlled trial . *Pediatrics* 2004 ;**114** :e1 –8.

Michelson D , Adler L L , Spencer T , et al. Atomoxetine in adults with ADHD: two randomized, placebo-controlled studies . *Biol Psychiatry* 2003 ;**53** (2):112–20 .

Michelson D , Buitelaar JK , Danckaerts M , et al. Relapse prevention in pediatric patients with ADHD treated with atomoxetine: a randomized, double-blind, placebo-controlled study . *J Am Acad Child Adolesc Psychiatry* 2004 ;**43** (7):896 –904 .

Stiefel G , Besag FM . Cardiovascular effects of methylphenidate, amphetamines, and atomoxetine in the treatment of attention-deficit hyperactivity disorder . *Drug Saf* 2010 ;**33** (10):821–42 .

Benztropine

Therapeutics

Brands

- Cogentin
- *see index for additional brand names*

Generic?

Yes

Class

- Antiparkinson agent; anticholinergic

Commonly Prescribed for

(bold for FDA approved)

- **Extrapyramidal disorders (drug-induced parkinsonism)**
- **Parkinsonism**
- Acute dystonic reactions
- Idiopathic generalized dystonia
- Focal dystonias
- Dopa-responsive dystonia

How the Drug Works

- Diminishes the excess acetylcholine activity caused by removal of dopamine inhibition when dopamine receptors are blocked
- May also inhibit the reuptake and storage of dopamine at central dopamine receptors, prolonging dopamine action

How Long Until It Works

- For drug-induced parkinsonism and in Parkinson's disease, onset of action can be within minutes or hours

If It Works

- Reduces motor side effects
- Does not lessen the ability of antipsychotics to cause tardive dyskinesia

If It Doesn't Work

- Consider switching to trihexyphenidyl, diphenhydramine, or a benzodiazepine
- Disorders that develop after prolonged antipsychotic use may not respond to treatment
- Consider discontinuing the agent that precipitated the parkinsonism

Best Augmenting Combos for Partial Response or Treatment Resistance

- If ineffective, switch to another agent rather than augment
- Benztropine is itself an augmenting agent to antipsychotics

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Prevents the action of acetylcholine on muscarinic receptors

Notable Side Effects

- Dry mouth, blurred vision, diplopia
- Confusion, hallucinations
- Constipation, nausea, vomiting
- Dilation of colon/paralytic ileus/bowel obstruction
- Erectile dysfunction



Life-Threatening or Dangerous Side Effects

- Angle-closure glaucoma
- Heat stroke, especially in elderly patients
- Tachycardia, cardiac arrhythmias, hypotension

- Urinary retention
- Anticholinergic agents such as benztropine can exacerbate or unmask tardive dyskinesia

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- For confusion or hallucinations, discontinue use
- For sedation, lower the dose and/or take the entire dose at night
- For dry mouth, chew gum or drink water
- For urinary retention, obtain a urological evaluation; may need to discontinue use

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Drug-induced parkinsonism: 2–8 mg/day
- Parkinsonism: 0.5–6 mg/day

Dosage Forms

- Tablet 0.5 mg, 1 mg, 2 mg
- Injection 1 mg/mL

How to Dose

- Drug-induced parkinsonism: 1–4 mg once or twice daily; can be given orally or parenterally
- Parkinsonism (oral): initial 0.5 mg once daily; increase by 0.5 mg at 5–6 day intervals until desired efficacy is reached



Dosing Tips

- If drug-induced parkinsonism occurs soon after initiation of a neuroleptic drug, it is likely to be transient; thus, attempt to withdraw benztropine after 1–2 weeks to determine if still needed
- Patients may take benztropine once daily at night to improve sleep and allow for earlier rising in the morning
- Taking benztropine with meals can reduce side effects
- Intramuscular and intravenous dosing are equally effective and fast-acting

Overdose

- Circulatory collapse, cardiac arrest, respiratory depression or arrest, psychosis, shock, coma, seizure, ataxia, combativeness, anhidrosis and hyperthermia, fever, dysphagia, decreased bowel sounds, sluggish pupils

Long-Term Use

- Safe
- Effectiveness may decrease over time (years), and side effects such as sedation and cognitive impairment may worsen

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Half-life 36 hours, although greatest effect lasts about 6–8 hours
- Metabolism is not well understood



Drug Interactions

- Use with amantadine may increase side effects

- Benztropine and all other anticholinergic agents may increase serum levels and effects of digoxin
- Can lower concentration of haloperidol and other phenothiazines, causing worsening of schizophrenia symptoms
- Can decrease gastric motility, resulting in increased gastric deactivation of levodopa and reduction in efficacy

Other Warnings/Precautions

- Use with caution in hot weather, as benztropine may increase susceptibility to heat stroke
- Anticholinergic agents have additive effects when used with drugs of abuse such as cannabinoids, barbiturates, opioids, and alcohol

Do Not Use

- In patients with glaucoma, particularly angle-closure glaucoma
- In patients with pyloric or duodenal obstruction, stenosing peptic ulcers, prostate hypertrophy or bladder neck obstructions, achalasia, or megacolon
- If there is a proven allergy to benztropine

Special Populations

Renal Impairment

- No known effects, but use with caution

Hepatic Impairment

- No known effects, but use with caution

Cardiac Impairment

- Use with caution in patients with known arrhythmias, especially tachycardia

Elderly

- Use with caution
- Elderly patients may be more susceptible to side effects



Children and Adolescents

- Do not use in children ages 3 and younger
- Generalized dystonias may respond to anticholinergic treatment, and young patients usually tolerate the medication better than the elderly
- Usual dose is 0.05 mg/kg once or twice daily



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women

Breast Feeding

- Unknown if benztropine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on benztropine should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- Drug-induced parkinsonism, particularly in acute setting

Potential Disadvantages

- Patients with long-standing drug-induced parkinsonism may not respond to treatment
- Generalized dystonias (less established as treatment than trihexyphenidyl)

Primary Target Symptoms

- Tremor, akinesia, rigidity, drooling

Pearls

- First-line agent for drug-induced parkinsonism related to antipsychotic use
- Useful adjunct in younger Parkinson's patients with tremor, but trihexyphenidyl is more commonly used
- Useful in the treatment of post-encephalitic Parkinson's disease and for drug-induced parkinsonism, but not for tardive dyskinesias
- Post-encephalitic Parkinson's patients usually tolerate higher doses better than idiopathic Parkinson's patients
- Generalized dystonias are more likely to benefit from anticholinergic therapy than focal dystonias; trihexyphenidyl is used more commonly than benztropine
- Sedation limits use, especially in older patients
- Patients with cognitive impairment may do poorly
- Can cause cognitive side effects with chronic use, so periodic trials of discontinuation may be useful to justify continuous use, especially in institutional settings when used as an adjunct to antipsychotics
- Can be abused in institutional or correctional settings
- Commonly used in an oral or intramuscular formulation as needed with concomitant antipsychotics to reduce or prevent drug-induced parkinsonism

Suggested Reading

Brocks DR . Anticholinergic drugs used in Parkinson's disease: an overlooked class of drugs from a pharmacokinetic perspective . *J Pharm Pharm Sci* 1999 ;**2** (2):39 –46 .

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Colosimo C , Gori MC , Inghilleri M . Postencephalitic tremor and delayed-onset parkinsonism . *Parkinsonism Relat Disord* 1999 ;**5** (3):123–4 .

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Costa J , Esp  rito-Santo C , Borges A , et al. Botulinum toxin type A versus anticholinergics for cervical dystonia . *Cochrane Database Syst Rev* 2005 ;(1):CD004312 .

.....

Hai NT , Kim J , Park ES , Chi SC . Formulation and biopharmaceutical evaluation of transdermal patch containing benztropine . *Int J Pharm* 2008 ;**357** (1–2):55 –60 .

Blonanserin

Therapeutics

Brands

- Lonasen
- *see index for additional brand names*

Generic?

No

Class

- Atypical antipsychotic (serotonin dopamine antagonist; second-generation antipsychotic; also a potential mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- Schizophrenia
- Acute mania/mixed mania
- Other psychotic disorders
- Bipolar maintenance
- Bipolar depression
- Treatment-resistant depression

- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognition and affective symptoms
- Actions at dopamine 3 receptors could theoretically contribute to blonanserin's efficacy

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients may require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms but does not eliminate them

- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine

- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, consider higher doses or augmentation with valproate or lamotrigine
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection
- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Mood-stabilizing anticonvulsants (carbamazepine, oxcarbazepine, lamotrigine)
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting any atypical antipsychotic